

Your first fsQCA, step by step

A complete worked example using a SmartPLS model of AAIR, HAIC, AIGM and BVC with 378 respondents. Every screenshot comes from SaNaSoft running on the study's latent variable scores.

SaNaSoft was developed by Prof. S. Sabraz Nawaz and Prof. Ghazanfar Ali

Time needed about 25 minutes **You need** your latent variable scores (CSV or Excel) and your model diagram

Software SaNaSoft, in any web browser

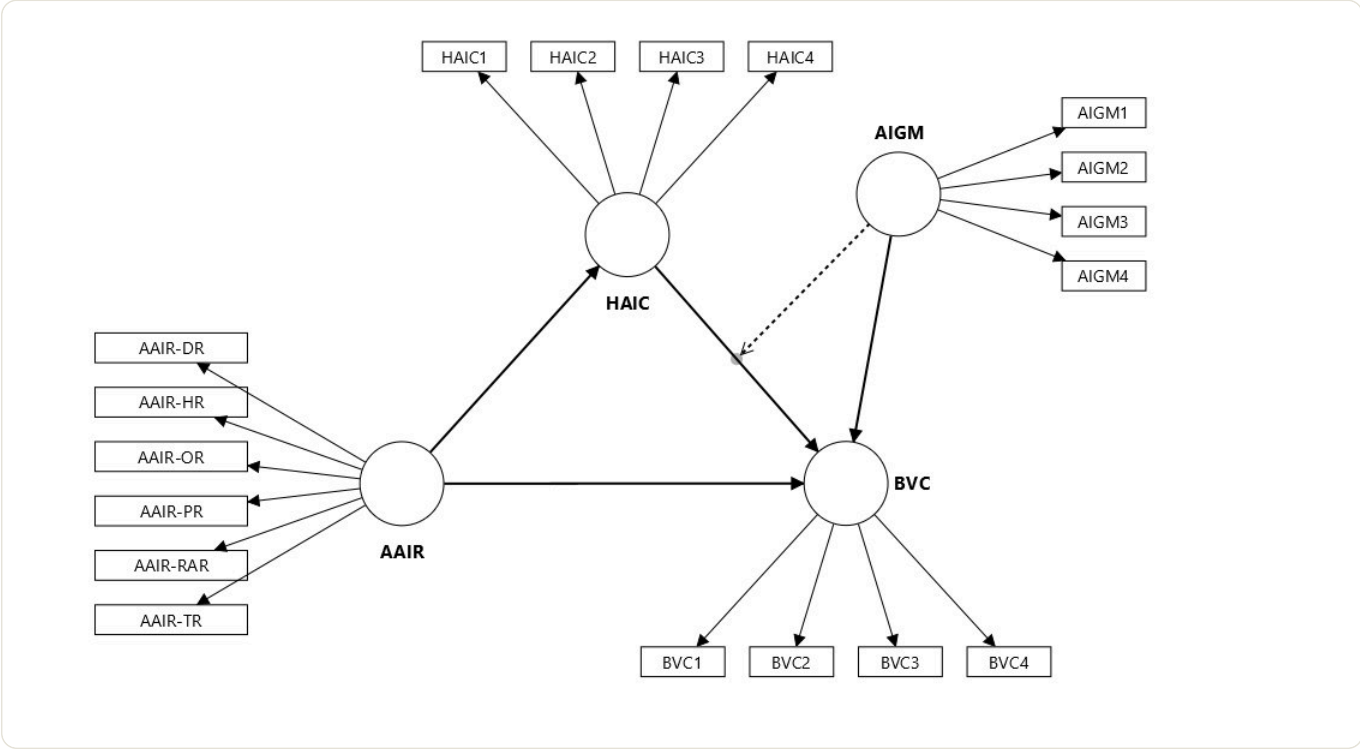
Contents

1. Before you start
2. Upload your scores
3. Check the variables
4. Describe your model
5. Convert scores to 0–1
6. Must-have factors
7. Combination table
8. Read your recipes
9. Check robustness and predictive validity
10. Download your results
11. Save your project and come back later
12. Run the remaining analyses
13. If you have raw items instead
14. Writing up the findings
15. How to cite SaNaSoft

Before you start

fsQCA asks a different question from SmartPLS. SmartPLS estimates how much each variable contributes *on average*. fsQCA asks which *combinations* of variables lead to a high (or low) result. SaNaSoft calls these combinations **recipes**; papers call them configurations.

This tutorial walks through a complete analysis of the model below, using its latent variable scores for 378 respondents (Book1.csv). Every screenshot comes from SaNaSoft 1.2 running on that file.



The SmartPLS model used in this tutorial. AAIR points to HAIC and BVC, HAIC points to BVC, and AIGM points to BVC and moderates the HAIC → BVC path (dotted arrow).

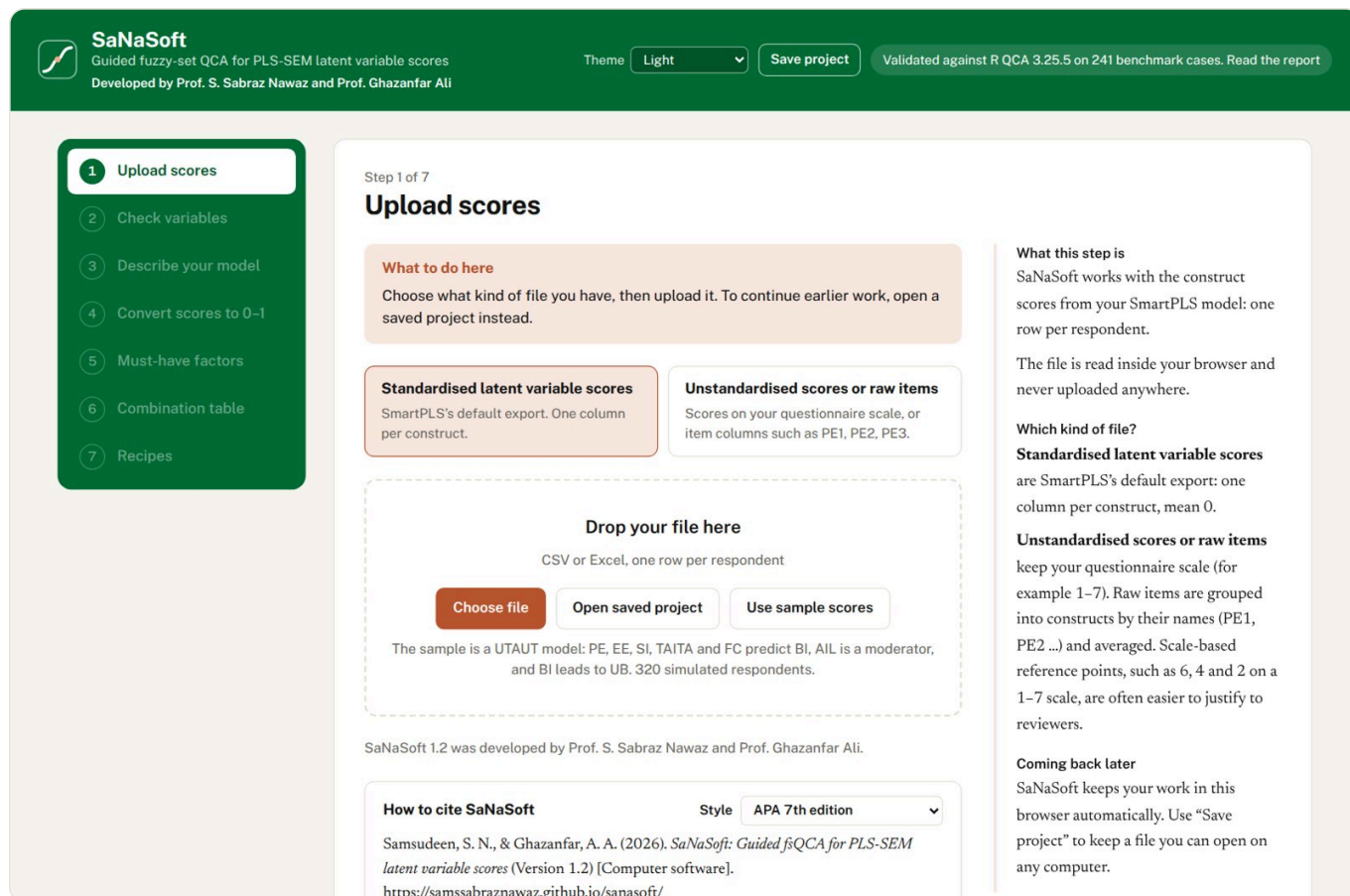
Here is how each part of the model is used in fsQCA:

In SmartPLS	Role in SaNaSoft	What happens in fsQCA
BVC	Dependent variable	The result you want to explain.
AAIR	Independent variable	A factor for HAIC and for BVC.
HAIC	Mediator	Analysed twice: as a result of AAIR, and as a factor for BVC.
AIGM	Moderator (also a direct arrow to BVC)	Included as an ordinary factor for BVC. No interaction term is needed.

Tip. Words with a dotted underline show the technical term used in papers when you point at them or tap them. The Theme menu at the top switches between Light, Dark and Match device.

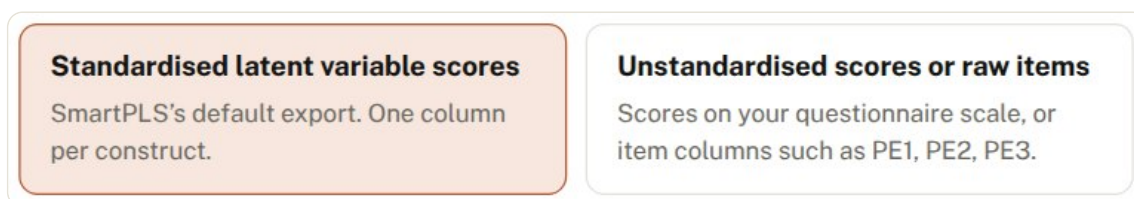
1 Upload your scores

Open SaNaSoft in any web browser. The green bar at the top shows the developers, a Theme menu and, once data are loaded, a **Save project** button. The green panel on the left lists the seven steps; the panel on the right explains each step in plain language.



The start screen of SaNaSoft 1.2.

First, choose what kind of file you have:



The two kinds of file SaNaSoft accepts.

- **Standardised latent variable scores:** SmartPLS's default export, one column per construct. Choose this for Book1.csv.
- **Unstandardised scores or raw items:** scores on your questionnaire scale, or item columns such as PE1, PE2, PE3. See section 12.

Then click **Choose file** (or drag the file onto the dashed box) and select Book1.csv. SaNaSoft confirms 378 respondents and 4 constructs. A row-number column or a SmartPLS moderation term such as "AIGM x HAIC" would be left out automatically.

What to do here

Choose what kind of file you have, then upload it. To continue earlier work, open a saved project instead.

Standardised latent variable scores

SmartPLS's default export. One column per construct.

Unstandardised scores or raw items

Scores on your questionnaire scale, or item columns such as PE1, PE2, PE3.

Drop your file here

CSV or Excel, one row per respondent

Choose file

Open saved project

Use sample scores

The sample is a UTAUT model: PE, EE, SI, TAITA and FC predict BI, AIL is a moderator, and BI leads to UB. 320 simulated respondents.

Loaded **Book1.csv**: 378 respondents and 4 constructs. Click Continue.

After uploading: a green message confirms the file and shows its first rows.

Below the upload box, SaNaSoft shows who developed it and how to cite it. Choose a reference style and click **Copy citation** to paste it into your paper.

How to cite SaNaSoft

Style **APA 7th edition**

Samsudeen, S. N., & Ghazanfar, A. A. (2026). *SaNaSoft: Guided fsQCA for PLS-SEM latent variable scores* (Version 1.2) [Computer software].

<https://samssabraznawaz.github.io/sanasoft/>

Copy citation

The "How to cite SaNaSoft" box. The style menu offers APA 7th, Harvard, Emerald, Chicago author-date, IEEE and Vancouver.

2 Check the variables

Each construct should have a mean of 0 and a standard deviation of 1, which confirms the scores are standardised. Yours are. Nothing needs changing, so click **Continue**.

Check variables

What to do here

Check that every construct from your SmartPLS model is listed. Usually you don't need to change anything: just click Continue.

Use	Variable	Mean	SD	Lowest	Highest	Missing
☒	AAIR	-0.00	1.00	-3.04	2.45	0
☒	AIMG	-0.00	1.00	-1.94	1.88	0
☒	BVC	0.00	1.00	-1.99	1.96	0
☒	HAIC	0.00	1.00	-2.00	1.89	0

What this step is

A quick check that SaNaSoft read your file correctly.

Standardised scores should each have a mean of 0 and a standard deviation of 1.

Raw items are grouped into constructs by name. Check the groups, fix any that are wrong, and list reverse-coded items so they are flipped before averaging. Cronbach's α is shown as a quick reliability check.

Higher-order constructs

If a construct is made of dimensions (for example four AI-literacy dimensions forming AIL), combine them. Its score is the mean of the dimension scores.

Columns left out

Row numbers and SmartPLS moderation terms (like "AIMG x HAIC") are left out automatically. You don't need the moderation term.

[Back](#)
[Continue to describe your model](#)

All four constructs have mean 0.00 and SD 1.00, with no missing values.

3 Describe your model

This is the only step that needs real input. Look at your SmartPLS diagram and choose each variable's role from the drop-down list:

1. Set **BVC** to **Dependent variable**.
2. Set **HAIC** to **Mediator**. Its arrow goes to BVC, which SaNaSoft ticks automatically.
3. Set **AAIR** to **Independent variable**. Leave both BVC and HAIC ticked, because AAIR has arrows to both.
4. Set **AIMG** to **Moderator**. Its dotted arrow points at the HAIC → BVC arrow, which is an arrow going into BVC, so keep **BVC** ticked and **untick HAIC**.

Variable	Role in your SmartPLS model	Paths
AAIR	Independent variable ▼	Arrows go to: BVC HAIC
AIGM	Moderator ▼	Its dotted arrow points at arrows going into: BVC HAIC
BVC	Dependent variable ▼	–
HAIC	Mediator ▼	Arrows go to: BVC

The model described in SaNaSoft. AIGM's arrow is ticked for BVC only.

What about AIGM's direct arrow to BVC? It makes no difference. Whether AIGM moderates an arrow into BVC or predicts BVC directly, fsQCA uses it the same way: as a factor for BVC. If AIGM matters, it appears in the recipes together with the factors it strengthens.

SaNaSoft now suggests two analyses based on your arrows:

Suggested analyses

Analysis 1 What leads to HAIC?

Result: **HAIC** · Factors: **AAIR**

HAIC is a mediator, so it is also a result of the variables pointing to it. This shows which combinations of them lead to high HAIC.

Only one factor. fsQCA needs at least two to find combinations, so this analysis adds little.

Start: what leads to high HAIC

What leads to low HAIC

Analysis 2 What leads to BVC?

Result: **BVC** · Factors: **AAIR, AIGM, HAIC**

Mirrors the arrows going into BVC in your SmartPLS model.

HAIC is included as a factor. If recipes combine it with a predictor, that supports the indirect route in your model.

AIGM is included as an ordinary factor. If it matters, it will appear in the recipes alongside the factors it strengthens.

Start: what leads to high BVC

What leads to low BVC

Report the analyses for both high and low levels: in fsQCA, the reasons for a low result are not simply the reverse of the reasons for a high result.

The suggested analyses. Analysis 2 (BVC) is the main one.

- **Analysis 1, what leads to HAIC?** Only AAIR points to HAIC, so this analysis has a single factor. SaNaSoft warns that fsQCA adds little with one factor.
- **Analysis 2, what leads to BVC?** This uses AAIR, AIGM and HAIC, mirroring every arrow into BVC. Click **Start: what leads to high BVC**.

4 Convert scores to 0–1

fsQCA asks how much each respondent belongs to the “high” group for each variable, from 0 (clearly low) to 1 (clearly high). For standardised scores SaNaSoft fills in the recommended reference points: the 95th, 50th and 5th percentiles of your data.

What to do here

The recommended settings are already filled in. Leave the tick box on and click Continue.

Only change the numbers if you have a clear reason. Click a variable's name to see its curve.

Use recommended: 95 / 50 / 5 percentiles

Robustness check: 90 / 50 / 10 percentiles

AAIR: from score to 0-1

Median score -0.00

In the high group 50%



The blue curve turns each score (along the bottom) into a 0-1 value (up the side).

Variable	Clearly high (0.95)	Middle point (0.5)	Clearly low (0.05)	Setting
AAIR	1.64	0	-1.65	95 / 50 / 5 percent
AIGM	1.72	-0.03	-1.62	95 / 50 / 5 percent
BVC	1.64	0.01	-1.68	95 / 50 / 5 percent
HAIC	1.58	0.08	-1.69	95 / 50 / 5 percent

☒ Keep respondents who sit exactly at the middle point (recommended)

Conversion formula

Same as fsQCA 4.x software, Ragin's method (recommended) ▼

The two differ only slightly: clearly high becomes 0.953 in fsQCA and 0.950 in R. Pick the one your field's papers usually cite.

Copy converted scores For checking the results in fsQCA 4.0 or R.

The blue curve shows how each BVC score becomes a 0-1 value; the grey bars show how respondents are spread.

What to do: nothing. Keep the recommended settings and the tick box on, and click **Continue**. The “Conversion formula” menu lets you match either fsQCA 4.x (the default) or the R QCA package.

5 Must-have factors

SaNaSoft checks whether high BVC ever occurs without a particular factor. None of your factors reaches the 0.90 benchmark (the highest is high AAIR at 0.771), so none is a must-have. This is common with survey data; report it in one sentence.

What leads to BVC? Now analysing what leads to **high BVC** , using 3 factors , N = 378.

Switch to low BVC

What to do here

Nothing to enter. Read the summary below, then click Continue.

None of the factors is a must-have for high BVC. This is common with survey data.
Report it in one sentence and continue.

Factor	Reliability	Share explained	Relevance	Verdict
High AAIR (AAIR)	0.771	0.773	0.814	Not a must-have
Low AAIR (~AAIR)	0.543	0.546	0.689	Not a must-have
High AIGM (AIGM)	0.712	0.716	0.780	Not a must-have
Low AIGM (~AIGM)	0.583	0.585	0.706	Not a must-have
High HAIC (HAIC)	0.755	0.766	0.814	Not a must-have
Low HAIC (~HAIC)	0.549	0.546	0.683	Not a must-have

None of the factors is a must-have for high BVC.

6 Combination table

Every respondent has a pattern of high and low values. With three factors there are eight possible patterns. SaNaSoft counts the respondents in each and checks whether the pattern reliably goes with high BVC.

What leads to BVC? Now analysing what leads to **high BVC**, using 3 factors, N = 378.

[Switch to low BVC](#)

What to do here

The recommended settings are already applied. Click Continue.

For your 378 respondents SaNaSoft recommends a minimum of **3** respondents per pattern, a reliability cutoff of **0.80** and a clean-signal cutoff of **0.70**.

Minimum respondents

Reliability cutoff

Clean-signal cutoff

[Use recommended settings](#)

Sort by

[Respondents](#) ▼

The patterns kept cover 378 of 378 respondents (100%, above the 80% target). 1 pattern leads reliably to high BVC.

AAIR	AIGM	HAIC	Respondents	Running total	Reliability	Clean signal	Lea
Low	Low	Low	81	21%	0.616	0.220	
High	High	High	78	42%	0.926	0.846	
Low	High	Low	51	56%	0.710	0.290	
High	Low	High	51	69%	0.847	0.616	
High	High	Low	35	78%	0.834	0.529	
Low	Low	High	31	87%	0.792	0.374	
Low	High	High	27	94%	0.869	0.586	
High	Low	Low	24	100%	0.810	0.419	

0 other patterns have no respondents at all. Rows marked "Too few" are set aside.

☒ **Block unrealistic assumptions** ([Enhanced Standard Analysis](#), recommended)

To simplify recipes, fsQCA makes assumptions about patterns with too few respondents. With this on, SaNaSoft refuses assumptions that would be used both for high and for low BVC (a contradiction), and assumptions that ignore a must-have factor. For this analysis it blocked 0 contradictory assumptions and 0 that ignore a must-have factor.

The combination table. Only one pattern, high AAIR with high AIGM and high HAIC (78 respondents, reliability 0.926), is marked "Yes".

What to do: nothing; the recommended settings are applied. For 378 respondents that means at least 3 respondents per pattern (keeping 100% of respondents), a reliability cutoff of 0.80 and a clean-signal cutoff of 0.70.

Notice the pattern with high AAIR, low AIGM and high HAIC: its reliability is 0.847 but its clean-signal score is only 0.616, so it does not count. It reappears in the robustness check. At the bottom, **Block unrealistic assumptions** (Enhanced Standard Analysis) is on; for this data it found nothing to block.

7 Read your recipes

At the top of the Recipes step, say what your theory expects. All SmartPLS paths into BVC are positive, so keep **Helps when high** for every factor.

What your theory expects

AAIR	Helps when high	Helps when low	Not sure
AIGM	Helps when high	Helps when low	Not sure
HAIC	Helps when high	Helps when low	Not sure

Theory expectations. Change a factor to “Helps when low” only if its SmartPLS path was negative.

The recipe chart shows one recipe for high BVC, and SaNaSoft explains it in plain words:

Recipes for high BVC		Copy table for your paper
Factor	Recipe 1	
AAIR		
AIGM		
HAIC		
Reliability	0.926	
Share explained	0.518	
Unique share	0.518	
Overall reliability	0.926	
Overall share explained	0.518	
 High, key ingredient  High, supporting ingredient  Low, key ingredient  Low, supporting ingredient Blank: doesn't matter		

Recipe 1: high AAIR, high AIGM and high HAIC, all key ingredients (large dots).

What this means

Recipe 1: high AAIR, high AIGM and high HAIC together lead to high BVC. Its reliability is 0.93 (above the 0.80 benchmark), and it accounts for about 52% of high BVC.

This recipe accounts for about 52% of high BVC, with overall reliability 0.93.

The same result in a sentence you can adapt for your paper.

In other words, **high BVC occurs when AAIR, AIGM and HAIC are all high together**. The recipe is highly reliable (0.93, well above 0.80) and accounts for about half of all high-BVC membership (0.52).

Share explained	0.518
Unique share	0.518
Overall reliability	0.926
Overall share explained	0.518

● High, key ingredient
● High, supporting ingredient
⊗ Low, key ingredient

⊗ Low, supporting ingredient

fsQCA papers call this a core condition: strong evidence it matters.

What this means

Recipe 1: high AAIR, high AIGM and high HAIC together lead to high BVC. Its reliability

Point at any dotted word to see its technical name: “key ingredient” is what papers call a core condition.

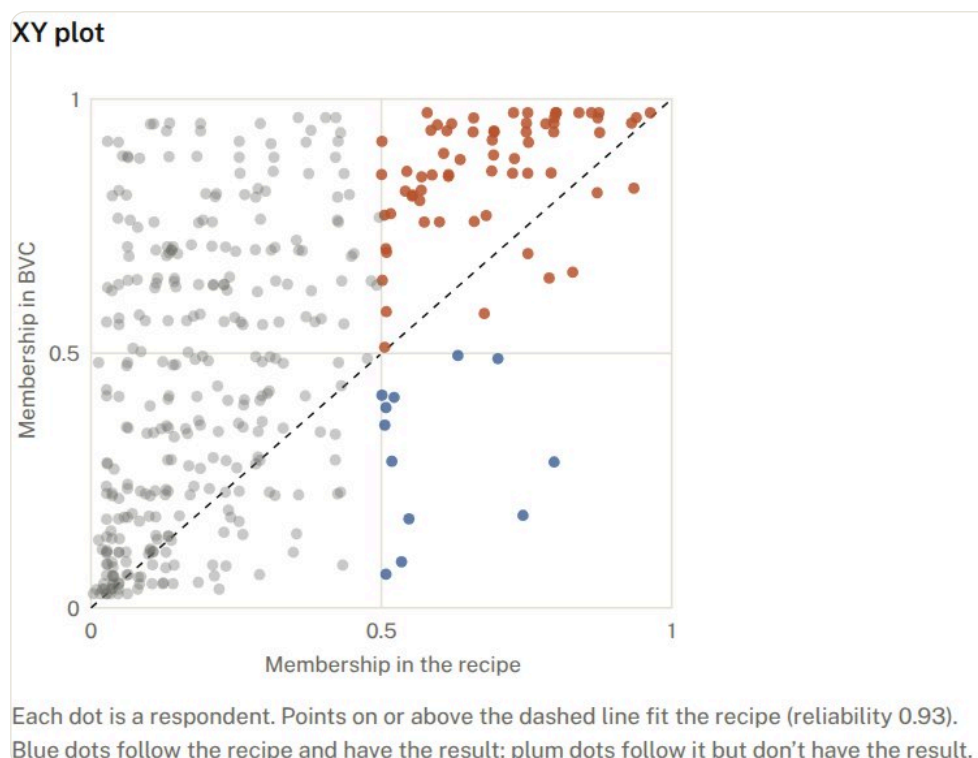
Further down, SaNaSoft shows which respondents follow the recipe, and an XY plot of the fit:

Who follows each recipe

A respondent “follows” a recipe when their membership in it is above 0.5. Exceptions are worth a look: they can suggest missing factors.

Recipe	Respondents who follow it	...and have high BVC	...but don't
Recipe 1 AAIR* AIGM* HAIC	78	66	

Respondents who follow the recipe and have high BVC, and exceptions who follow it but don't.



Each dot is a respondent. Dots on or above the dashed line fit the recipe.

8 Check robustness and predictive validity

Click **Run robustness check**. SaNaSoft reruns the analysis with other reference points, cutoffs and minimum numbers of respondents.

Robustness checkRun again

Reruns your analysis with different reference points and cutoffs, and shows whether the recipes stay the same. Reviewers increasingly expect this.

Check	Recipes found	Overall reliability	Overall share exp
Your settings	AAIR*AIMG*HAIC	0.926	C
Reference points 90 / 50 / 10	AAIR*AIMG*HAIC	0.914	C
Reference points 80 / 50 / 20	AAIR*AIMG*HAIC	0.902	C
Reliability cutoff 0.75	AAIR*AIMG*HAIC	0.926	C
Reliability cutoff 0.85	AAIR*AIMG*HAIC	0.926	C
Clean-signal cutoff 0.60	AAIR*HAIC	0.857	C
Clean-signal cutoff 0.75	AAIR*AIMG*HAIC	0.926	C
Minimum respondents 4	AAIR*AIMG*HAIC	0.926	C
Minimum respondents 2	AAIR*AIMG*HAIC	0.926	C

7 of 8 checks give exactly the same recipes. Where recipes change, look at whether the key ingredients stay the same; small changes are normal. Report this table and discuss the differences briefly.

7 of 8 checks give exactly the same recipe.

The one change is instructive. With a lower clean-signal cutoff (0.60), the recipe becomes **AAIR·HAIC** without AIGM. So the robust core is that **AAIR and HAIC together** lead to high BVC, and AIGM completes the recipe under the stricter, recommended setting.

Then click **Run predictive test**. SaNaSoft finds the recipe in a random half of the respondents and tests it on the other half.

Predictive validity test

[Run again](#)

Splits your respondents randomly into two halves, finds the recipes in the first half, and checks how well they predict the result in the second half.

Recipe found in half A (189 respondents)	Reliability in half B	Share explained in half B
AAIR*AIGM*HAIC	0.925	0.516
All recipes together	0.925	0.516

The recipes found in one half of your respondents also hold in the other half (reliability 0.92, above 0.80). This supports their predictive validity (Pappas & Woodside, 2021).

The recipe holds in the unseen half (reliability 0.925, share explained 0.516).

9 Download your results

The **Download your results** section turns everything into files for your paper. Choose the paper size, the **reference style** (APA 7th, Harvard, Emerald, Chicago author-date, IEEE or Vancouver) and the font, tick what to include, then download.

Download your results

Choose a reference style for the citations and reference list. Tables and figures follow APA 7th edition style: numbered tables and figures with italic titles, horizontal rules only, notes below each table, and no leading zeros for proportions. Tables open as real Word tables you can edit.

Paper size

US Letter ▼

Reference style

APA 7th edition ▼

Font

Times New Roman 12 pt ▼

Methods paragraph

Calibration table

Must-have table

Combination table

Recipe table

Robustness table

Predictive table

XY plots

Plain-language summary

Download Word report (APA 7)

Download Excel workbook

Save project

The download section. Robustness and predictive tables become available once those checks have been run.

Word report (APA 7). Tables and figures follow APA 7th edition style: numbered, with italic titles, horizontal rules only, notes below and no leading zeros for proportions. They are real Word tables you can edit. Citations and the reference list follow the style you chose, and SaNaSoft is cited automatically.

Fuzzy-Set Qualitative Comparative Analysis of High BVC

Results generated with SaNaSoft 1.2 (Samsudeen & Ghazanfar, 2026) on 24 September 2026

Method

Fuzzy-Set Qualitative Comparative Analysis

We conducted fsQCA on 378 respondents in SaNaSoft (Samsudeen & Ghazanfar, 2026), using the latent variable scores from the PLS-SEM model as construct scores. Because the latent variable scores were standardised, all constructs were calibrated using the direct method (Ragin, 2008) with the 95th, 50th and 5th sample percentiles as thresholds for full membership, the crossover point and full non-membership. Membership scores of exactly .5 were recoded to .501 to avoid dropping cases at the crossover. The truth table for BVC included 3 conditions (AAIR, AIGM and HAIC). Using a frequency threshold of 3, which retained 100% of cases, a raw consistency threshold of .80 and a PRI threshold of .70, 1 configuration was coded as sufficient. Following the Enhanced Standard Analysis (Schneider & Wagemann, 2012), contradictory simplifying assumptions (0) and remainders contradicting a necessary condition (0) were excluded from the minimisation. The intermediate solution, based on the directional expectations that AAIR, AIGM and HAIC would contribute to BVC when present, comprises 1 configuration with an overall solution consistency of .926 and solution coverage of .518.

Results

No condition was necessary for BVC: all necessity consistencies were below .90.

The intermediate solution for BVC comprised 1 configuration (solution consistency = .926, solution coverage = .518). Configuration 1 (AAIR*AIGM*HAIC) had a consistency of .926, raw coverage of .518 and unique coverage of .518.

Table 4*Configurations Sufficient for High BVC*

Condition	Configuration 1
AAIR	●
AIGM	●
HAIC	●
Consistency	.926
Raw coverage	.518
Unique coverage	.518
Solution consistency	.926
Solution coverage	.518

Note. Large filled circles (●) indicate core conditions present; small filled circles indicate peripheral conditions present; large crossed circles (⊗) indicate core conditions absent; small crossed circles indicate peripheral conditions absent; blank cells indicate that the condition may be either present or absent (Fiss, 2011). Intermediate solution; core conditions appear in both the parsimonious and intermediate solutions. Frequency threshold = 3; raw consistency threshold = .80; PRI threshold = .70.

Table 5*Robustness Checks for High BVC*

Specification	Solution	Consistency	Coverage	Same
Your settings	AAIR*AIGM*HAIC	.926	.518	Yes
Reference points 90 / 50 / 10	AAIR*AIGM*HAIC	.914	.481	Yes
Reference points 80 / 50 / 20	AAIR*AIGM*HAIC	.902	.404	Yes
Reliability cutoff 0.75	AAIR*AIGM*HAIC	.926	.518	Yes
Reliability cutoff 0.85	AAIR*AIGM*HAIC	.926	.518	Yes
Clean-signal cutoff 0.60	AAIR*HAIC	.857	.650	No
Clean-signal cutoff 0.75	AAIR*AIGM*HAIC	.926	.518	Yes
Minimum respondents 4	AAIR*AIGM*HAIC	.926	.518	Yes
Minimum respondents 2	AAIR*AIGM*HAIC	.926	.518	Yes

Note. Each row reruns the analysis with one setting changed. "Same" indicates an intermediate solution identical to the main analysis.

Page 3 of the Word report: the configuration table and robustness checks in APA 7 format.

Excel workbook. Nine sheets: settings and anchors, calibrated data, necessity, truth table, all three solutions, configuration chart, respondents per recipe, robustness and predictive validity.

Methods paragraph. The Recipes step also shows a ready-to-edit methods paragraph with its references in your chosen style. **Copy paragraph** copies the references too.

Paragraph for your methods section

Copy paragraph

We conducted fsQCA on 378 respondents in SaNaSoft (Samsudeen & Ghazanfar, 2026), using the latent variable scores from the PLS-SEM model as construct scores. Because the latent variable scores were standardised, all constructs were calibrated using the direct method (Ragin, 2008) with the 95th, 50th and 5th sample percentiles as thresholds for full membership, the crossover point and full non-membership. Membership scores of exactly 0.5 were recoded to 0.501 to avoid dropping cases at the crossover. The truth table for BVC included 3 conditions (AAIR, AIGM and HAIC). Using a frequency threshold of 3, which retained 100% of cases, a raw consistency threshold of 0.80 and a PRI threshold of 0.70, 1 configuration was coded as sufficient. Following the Enhanced Standard Analysis (Schneider & Wagemann, 2012), contradictory simplifying assumptions (0) and remainders contradicting a necessary condition (0) were excluded from the minimisation. The intermediate solution, based on the directional expectations that AAIR, AIGM and HAIC would contribute to BVC when present, comprises 1 configuration with an overall solution consistency of 0.926 and solution coverage of 0.518.

References

Ragin, C. C. (2008). *Redesigning social inquiry: Fuzzy sets and beyond*. University of Chicago Press.

Samsudeen, S. N., & Ghazanfar, A. A. (2026). *SaNaSoft: Guided fsQCA for PLS-SEM latent variable scores* (Version 1.2) [Computer software]. <https://samssabraznawaz.github.io/sanasoft/>

Schneider, C. Q., & Wagemann, C. (2012). *Set-theoretic methods for the social sciences: A guide to qualitative comparative analysis*. Cambridge University Press.

Papers use the technical terms, so this paragraph does too. Citations follow the reference style chosen under "Download your results" (now: APA 7th edition). "Copy paragraph" copies the references too.

The methods paragraph, citing SaNaSoft and the methodological sources in APA style.

Every download is numbered so nothing is ever overwritten:

File	First download	Next download
Word report	SaNaSoft_1_Report_Book1_high_BVC.docx	SaNaSoft_2_Report_Book1_high_BVC.docx
Excel workbook	SaNaSoft_1_Results_Book1_high_BVC.xlsx	SaNaSoft_2_Results_Book1_high_BVC.xlsx
Project	SaNaSoft_1_Project_Book1.sanasoft.json	SaNaSoft_2_Project_Book1.sanasoft.json

10 Save your project and come back later

Click **Save project** in the green bar (or in the download section) at any time. The project file keeps your data, model, settings and choices, and it opens on any computer: on the first screen click **Open saved project** and choose the file. SaNaSoft returns to the step where you saved, with identical results.

**SaNaSoft**
Guided fuzzy-set QCA for PLS-SEM latent variable scores
Developed by Prof. S. Sabraz Nawaz and Prof. Ghazanfar Ali

Theme Light

Save project

Validated against R QCA 3.25.5 on 241 benchmark cases. [Read the report](#)

The green bar: Theme menu, Save project and the link to the validation report.

SaNaSoft also keeps your work in the browser automatically. If you close or reload the page, the first screen offers to continue:

You have unfinished work from 9/24/2026, 1:44:26 AM: **Book1.csv**.

Continue where you left off

Start fresh

After reloading, SaNaSoft offers to continue where you left off.

11

Run the remaining analyses

At the bottom of the Recipes step, buttons run your other analyses with the same settings.

Run your other analyses

Your settings carry over. Each button runs straight through to the recipes; check steps 5 and 6 for each one too.

What leads to high HAIC

What leads to low HAIC

What leads to low BVC

Your other analyses: low BVC, and high and low HAIC.

Low BVC. The recipe is the mirror image: low AAIR, low AIGM and low HAIC together lead to low BVC (reliability 0.868, share explained 0.507).

Recipes for low BVC

Copy table for your paper

Factor	Recipe 1
AAIR	⊗
AIGM	⊗
HAIC	⊗
Reliability	0.868
Share explained	0.507
Unique share	0.507
Overall reliability	0.868
Overall share explained	0.507

● High, key ingredient

● High, supporting ingredient

⊗ Low, key ingredient

⊗ Low, supporting ingredient

Blank: doesn't matter

The recipe for low BVC. Crossed circles mean the factor must be low.

HAIC. With AAIR as its only factor, no pattern passes the settings, so SaNaSoft reports no recipe for high HAIC. That is a legitimate finding: AAIR alone does not reliably produce high HAIC. Report it rather than lowering the cutoffs to force a result.

12 If you have raw items instead

If your file holds questionnaire items (for example HAIC1 to HAIC4 on a 1–7 scale) or unstandardised scores, choose **Unstandardised scores or raw items** in step 1. SaNaSoft then:

- groups items into constructs by name and averages them, showing Cronbach's α ;
- flips reverse-coded items that you list;
- builds higher-order constructs, for example AAIR from its dimensions;
- uses scale-based reference points, such as 6 (agree), 4 (neutral) and 2 (disagree) on a 1–7 scale.

What to do here

Check that each construct has the right items. Set the scale, and list any reverse-coded items. Then click Continue.

Left out automatically: **RespID** (respondent ID).

Scale lowest point Scale highest point Reverse-coded items

1 7 HAIC4

Use	Variable	Items (comma-separated)	Type	Mean	SD
☒	AAIRDR	AAIR-DR1, AAIR-DR2, AAIR	Item mean	4.35	1.12
☒	AAIRHR	AAIR-HR1, AAIR-HR2, AAIF	Item mean	4.38	1.14
☒	AAIROR	AAIR-OR1, AAIR-OR2, AAIF	Item mean	4.23	1.16
☒	HAIC	HAIC1, HAIC2, HAIC3, HAIC4	Item mean	4.35	1.22
☒	AIGM	AIGM1, AIGM2, AIGM3, AIGM4	Item mean	4.34	1.16
☒	BVC	BVC1, BVC2, BVC3, BVC4	Item mean	4.21	1.13

Higher-order constructs

Name

AAIR Remove

AAIRDR AAIRHR AAIROR HAIC AIGM BVC

Add higher-order construct

Step 2 with an example raw-item file: HAIC4 is reverse-coded, and AAIR is built from three dimensions.

The rest of the analysis works exactly as described above, and the methods paragraph describes the scale-based calibration.

Writing up the findings

Here is how the results could be reported alongside the PLS-SEM findings. Adapt the wording to your paper.

fsQCA was conducted in SaNaSoft (Samsudeen & Ghazanfar, 2026). No single condition was necessary for high BVC (all consistencies below .90). The intermediate solution contained one configuration: high AAIR combined with high AIGM and high HAIC, all as core conditions (consistency = .926, coverage = .518). The negated outcome showed the reverse configuration, with low AAIR, low AIGM and low HAIC leading to low BVC (consistency = .868, coverage = .507). The solution was stable across alternative calibration anchors, consistency thresholds and frequency thresholds (7 of 8 robustness checks); with a PRI threshold of .60, AIGM dropped out, leaving AAIR*HAIC (consistency = .857, coverage = .650). A holdout test supported predictive validity (consistency = .925 in the holdout subsample).

What it adds to the PLS-SEM results:

- **Mediation.** AAIR and HAIC appear together in every recipe, consistent with AAIR working through HAIC. High AAIR on its own is not enough.
- **Moderation.** AIGM completes the recipe for high BVC, the configurational counterpart of AIGM strengthening the HAIC → BVC path.
- **Asymmetry.** With only three factors, the recipe for low BVC mirrors the one for high BVC. Say so, because it shows you checked.

How to cite SaNaSoft

SaNaSoft was developed by Prof. S. Sabraz Nawaz and Prof. Ghazanfar Ali. Please cite it in any work that uses it, for example:

Samsudeen, S.N. and Ghazanfar, A.A. (2026), "SaNaSoft: Guided fsQCA for PLS-SEM latent variable scores", Version 1.2, available at: <https://samssabraznawaz.github.io/sanasoft/>

The "How to cite SaNaSoft" box on the first screen gives the citation in six reference styles, and every report SaNaSoft produces includes it automatically.

Checklist before you submit

- ☐ Calibration anchors reported, with the .501 adjustment
- ☐ Necessity results reported, even if no condition was necessary
- ☐ Frequency, consistency and PRI thresholds reported, with the share of cases retained
- ☐ Configuration table with core and peripheral conditions, consistency and coverage
- ☐ Negated outcome analysed and reported
- ☐ Robustness table in an appendix; predictive validity test mentioned
- ☐ SaNaSoft cited in your chosen reference style

SaNaSoft 1.2 was developed by Prof. S. Sabraz Nawaz and Prof. Ghazanfar Ali. It is **validated against R QCA 3.25.5**. Tutorial updated September 2026.